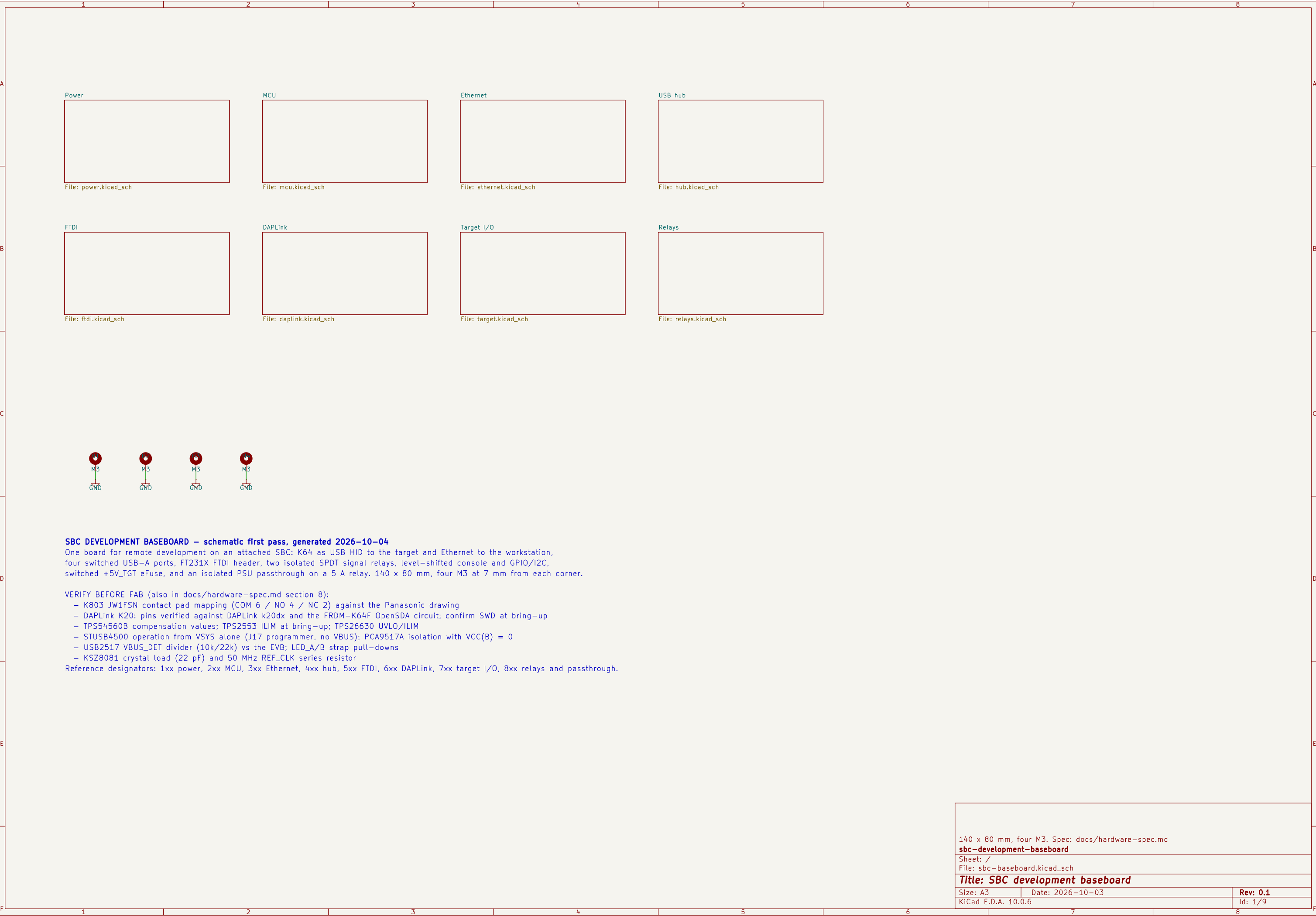
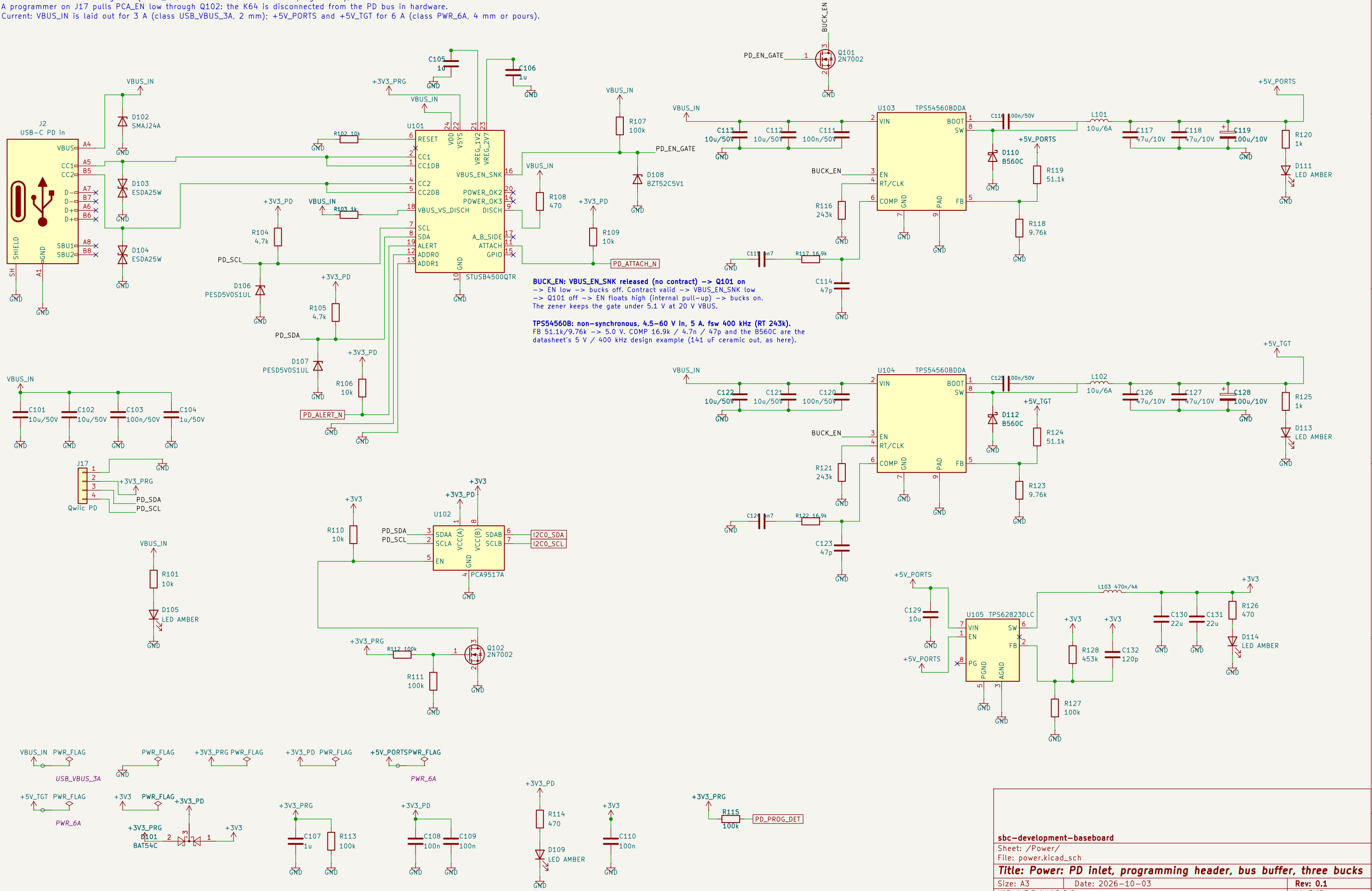


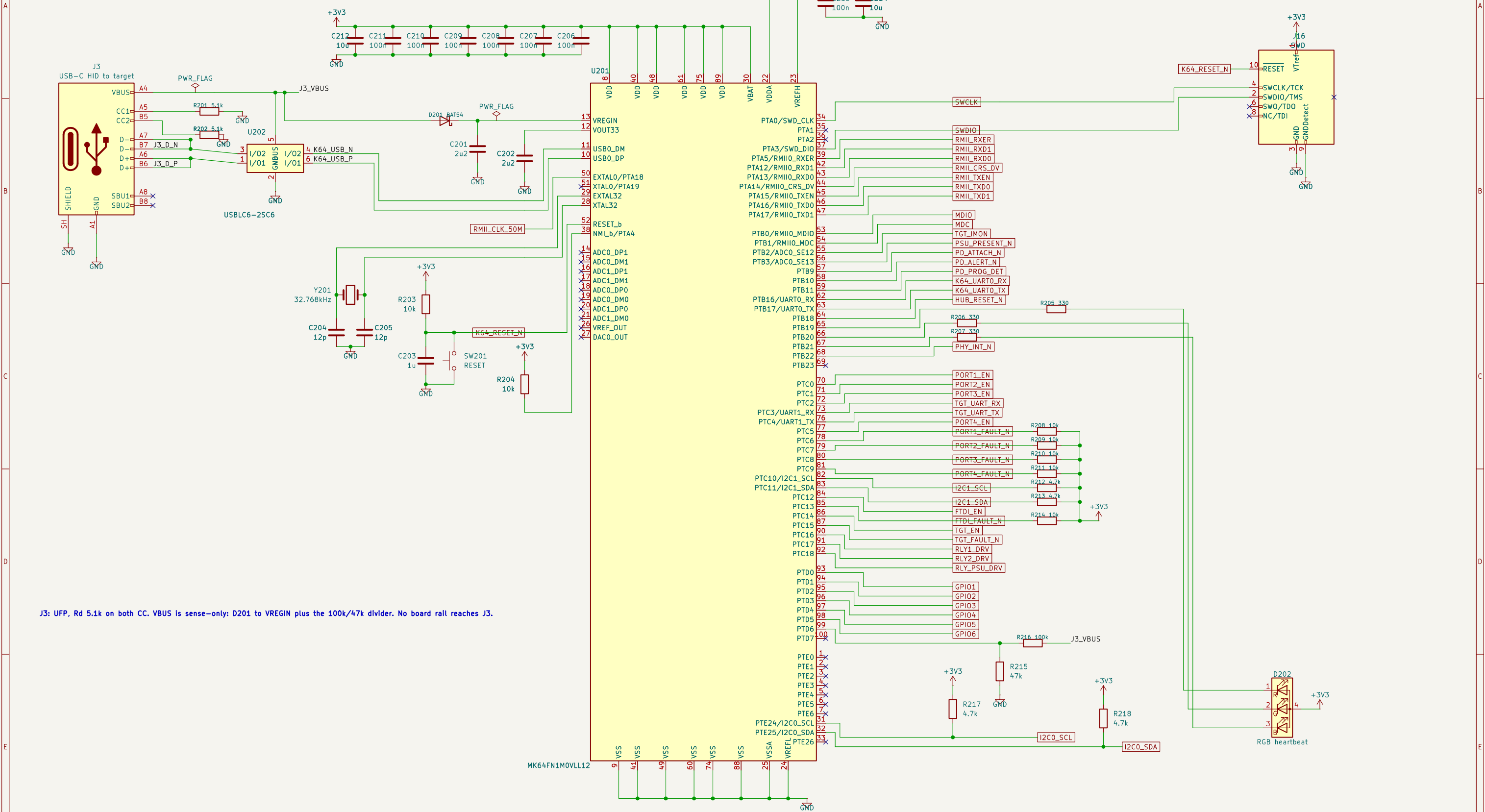


POWER

J2 is a PD sink only (no data). STUSB4500 negotiates from NVM; VBUS_EN_SNK gates the 5 V bucks through Q101, so the rails only come up once a sink contract (or plain Type-C 5 V) is valid. No series VBUS FET.
J17 (Qwiic) programs the STUSB4500 NVM and powers it through VSYS when the board is dead.
PCA9517A: A side = PD segment (+3V3_PD, diode-OR of +3V3 and the programmer), B side = K64 I2C0.
A programmer on J17 pulls PCA_EN low through Q102: the K64 is disconnected from the PD bus in hardware.
Current: VBUS_IN is laid out for 3 A (class USB_VBUS_3A, 2 mm); +5V_PORTS and +5V_TGT for 6 A (class PWR_6A, 4 mm or pours).

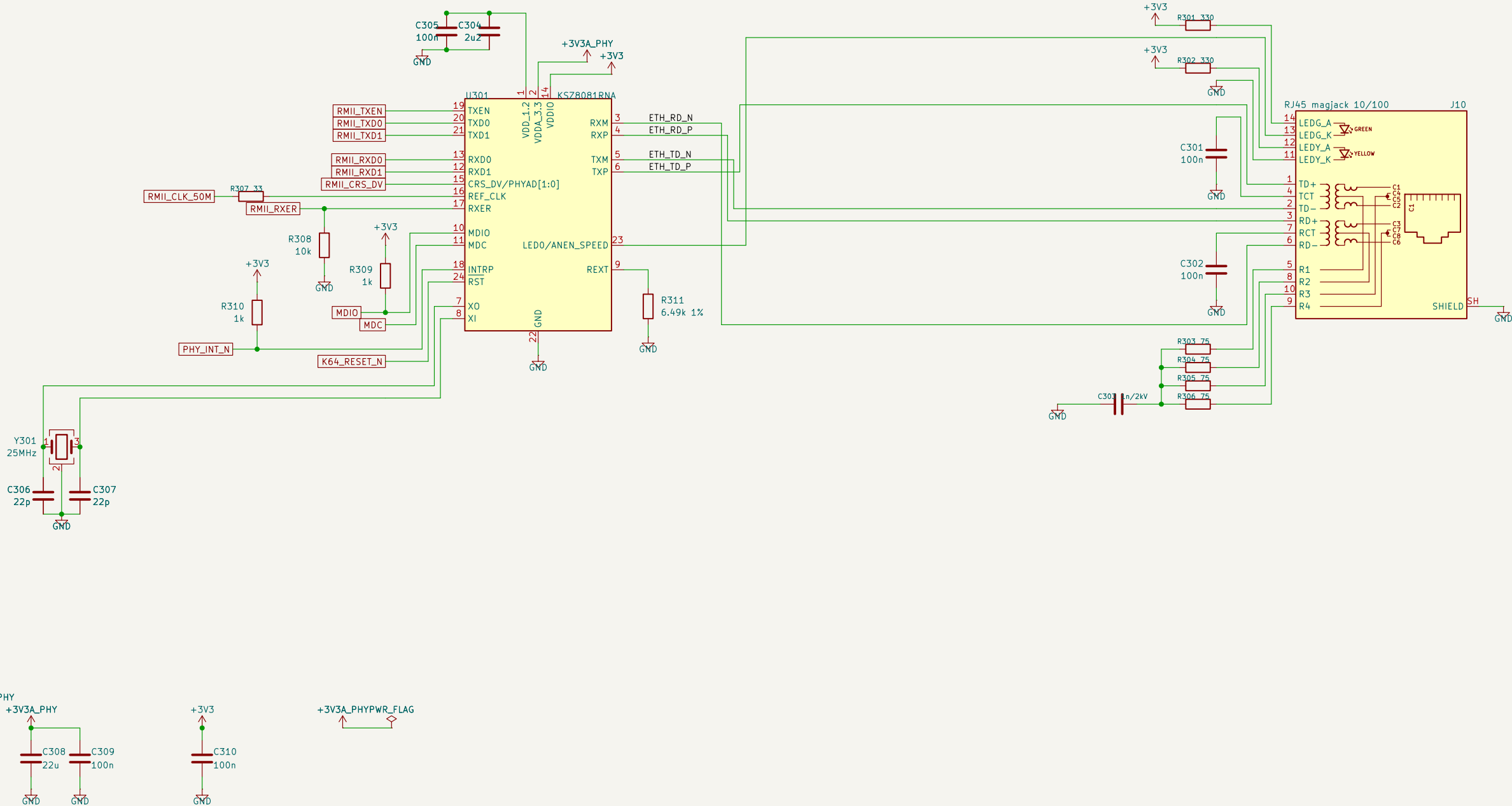


MCU
MK64FN1M0VLL12 (LQFP-100). Clock: 50 MHz RMII REF_CLK from the KSZ8081 into EXTAL0, as on the FRDM-K64F.
USB regulator: VREGIN fed only from J3 VBUS through D201, so the K64 cannot back-feed the target and the D+ pull-up disappears when the target is off. VOUT33 powers the transceiver. Pin allocation in docs/hardware-spec.md.
Spare pins (PTA1, PTA2, PTB23, PTD7, PTE0-6, PTE26, ADC, DAC) are left no-connect.
USB 2.0: K64_USB_P/N (K64 to U202) and J3_D_P/N (U202 to J3) are 90 ohm differential pairs, net class USB: route as pairs, no stubs.



J3: UFP, Rd 5.1k on both CC. VBUS is sense-only: D201 to VREGIN plus the 100k/47k divider. No board rail reaches J3.

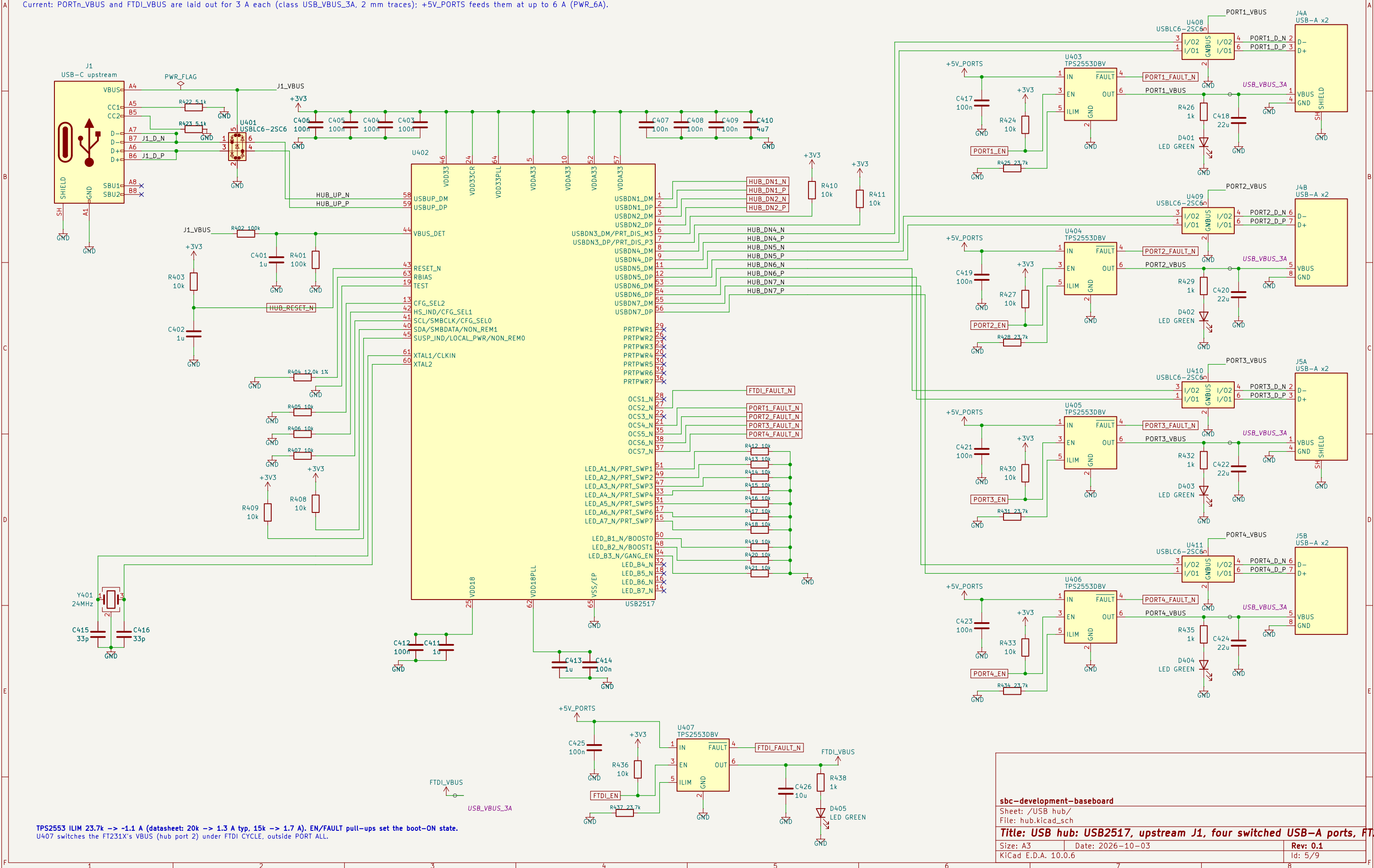
ETHERNET
KSZ8081RNA in RMII 25 MHz mode (its power-up default): 25 MHz crystal on XI/XO, REF_CLK outputs 50 MHz to the K64 EXTAL0 through R301. This is the FRDM-K64F arrangement (Zephyr: microchip,interface-type = rmii-25MHz). PHY address 0 (pins 15/16 internal pull-downs; the MAC does not drive them). RXER needs the external pull-down. Transformer centre taps go to 0.1 uF each, not to a supply (datasheet 11.x). Bob Smith: 4x75R to 1 nF/2 kV.



Crystal: 25 MHz +/-50 ppm, CL 16 pF typ (datasheet) -> 22 pF each, trim at bring-up.
Green LED = link/activity (PHY LED0, active low). Yellow LED = PHY powered (cathode to GND).

USB HUB

USB2517 strap-configured (CFG_SEL=000), no firmware needed for the USB tree to come up.
Port map: 1 = DAPLink, 2 = FT231X, 3 = disabled (DN3 pulled up), 4..7 = USB-A PORT1..PORT4 (J4A/J4B = PORT1/PORT2 on hub ports 4,5; J5A/J5B = PORT3/PORT4 on ports 6,7).
NON_REM=11 (pin 40 up, pin 45 up) marks ports 1-3 non-removable and LOCAL_PWR high reports self-powered.
LED_A/B pins pulled down: PRT_SWP normal polarity, BOOST=00, GANG_EN=0 (individual over-current).
Port power is switched by TPS2553 under K64 GPIO (boot ON via pull-ups); PRT_PWR outputs unused; /FAULT feeds OCSx_N.
USB 2.0 pairs are named *_P / *_N (KiCad's differential-pair convention) and sit in net class USB: 90 ohm differential, routed as pairs, no stubs. Each ESD array splits its pair into a hub side (HUB_DNn) and a connector side (PORTn_D). Set the class's width and gap from the stackup before routing.
Current: PORTn_VBUS and FTDI_VBUS are laid out for 3 A each (class USB_VBUS_3A, 2 mm traces); +5V_PORTS feeds them at up to 6 A (PWR_6A).

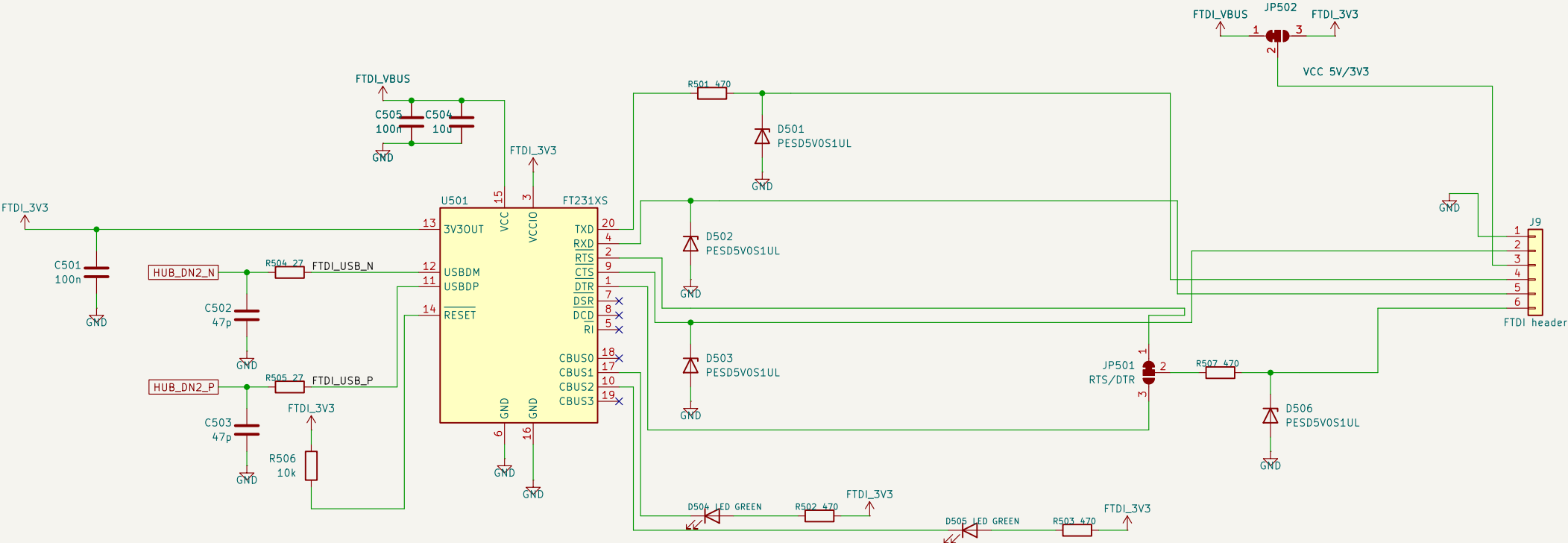


TPS2553 ILIM 23.7k -> -1.1 A (datasheet: 20k -> 1.3 A typ, 15k -> 1.7 A). EN/FAULT pull-ups set the boot-ON state.
U407 switches the FT231X's VBUS (hub port 2) under FTDI CYCLE, outside PORT ALL.

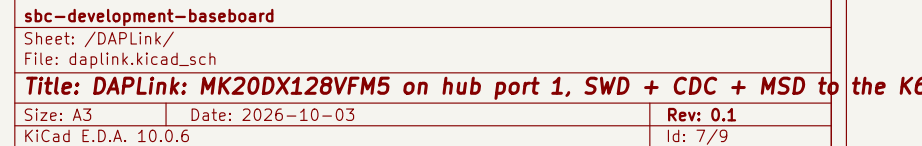
sbc-development-baseboard

Sheet: /USB hub/ File: hub.kicad_sch		
Title: USB hub: USB2517, upstream J1, four switched USB-A ports, FT231X s		
Size: A3	Date: 2026-10-03	Rev: 0.1
KiCad E.D.A. 10.0.6	Id: 5/9	

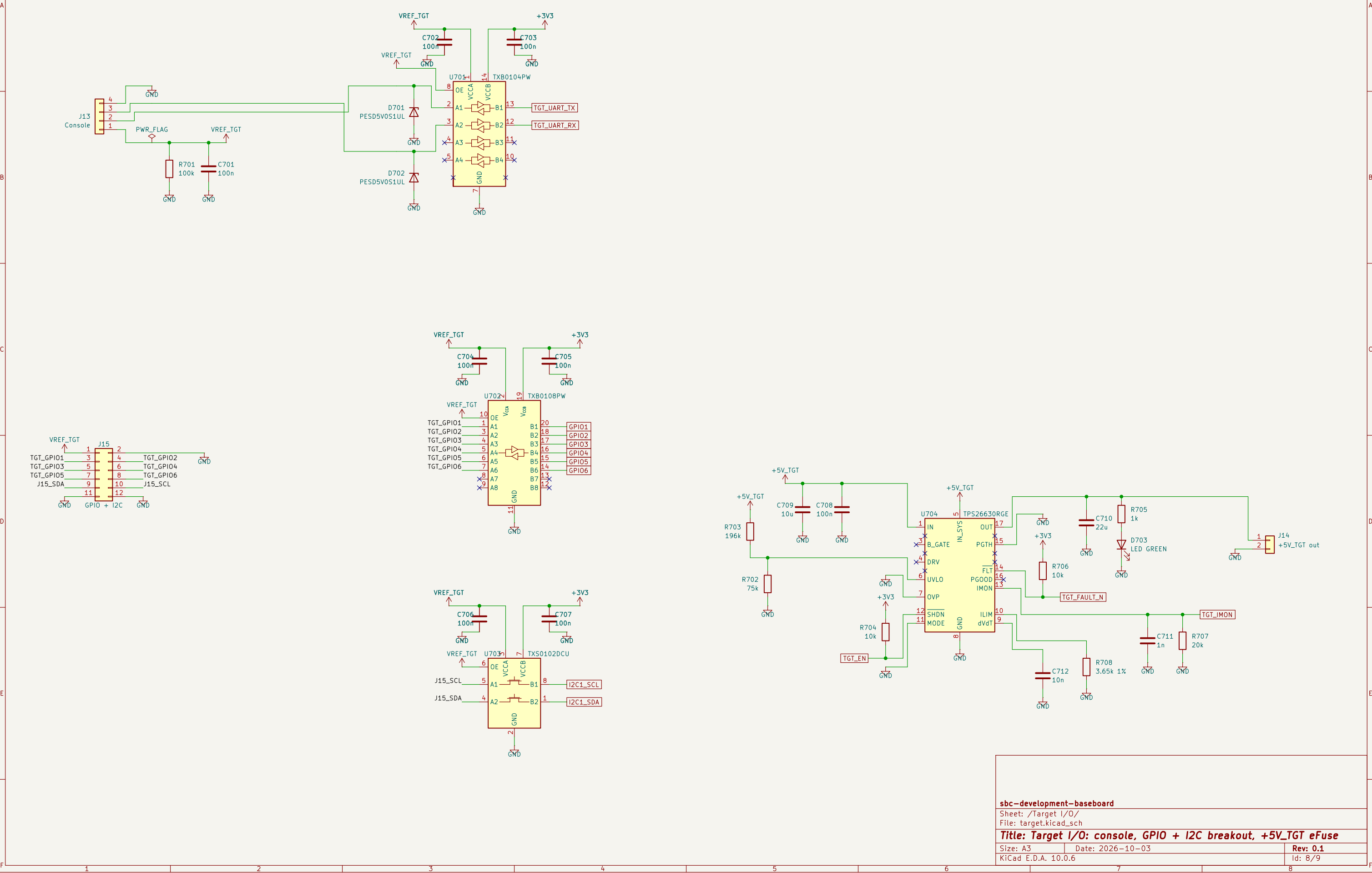
FTDI HEADER J9
Pinout = Adafruit FTDI Friend (the TTL-232R order): 1 GND (blk) 2 CTS (brn) 3 VCC (red) 4 TXD (org) 5 RXD (yel) 6 RTS/DTR (grn).
TXD/RXD named from the FT231X side: TXD drives the target's RX. 3.3 V levels only; 1.8 V consoles use J13.
JP501: pin 6 = RTS# (A-C, bridged by default) or DTR# (cut, bridge B-C). JP502: pin 3 VCC = FTDLVBUS 5 V (A-C, default) or 3V3OUT 3.3 V / 50 mA (B-C).
VCC comes from the TPS2553 on the hub sheet (FTDLVBUS): FTDI CYCLE power-cycles the bridge without a replug.
CBUS defaults: CBUS1 = RXLED#, CBUS2 = TXLED# (factory MTP); change with FT_PROG if ever needed.



MK20DXX128VFM5 running DAPLink (k20dx HCTC), powered from +5V_PORTS through its own USB regulator (VOUT33 -> VDD).
 Pin 20 provides the DAPLink k20dx HCTC: PTC5 = SWCLK, PTC6 = SWDIO, PTB1 = nRESET to the K64, PTD4 = LED.
 UART1 PTC3/PTC4 bridges to the K64 UART0 (CDC console). Pins verified against DAPLink k20dx IQ_Config.h and the FRDM-K64F OpenSDA circuit:
 PTC7 (SWDIO_IN) ties to SWDIO, 33 R in series with USB D+/D- (FRDM R20/R22). No level shifters: the K64 target is 3.3 V like the K20.
 J601 provides the K20 itself (bootloader + DAPLink). SW601 held at power-up enters DAPLink maintenance mode.



TARGET-FACING I/O
VREF_TGT comes from the target on J13.1 / J15.1 and sets the level of every translator (TXB0104/TXB0108/TXS0102).
OE pins follow VREF_TGT: with the target off the translators are high-impedance. J15 GPIO are push-pull (TXB).
TXS0102 instead of PCA9306: works with VREF_TGT equal to 3.3 V (PCA9306 needs VREF2 > VREF1). Range 1.65–3.6 V.
TPS26630: ILIM 3.65k -> -5 A, UVLO 196k/75k -> 4.3 V, OVP/MODE/PGTH to GND (latch-off). IMON 20k -> 2.8 V at 5 A into PTB2.



RELAYS AND PASSTHROUGH

K801/K802 G6K-2F-Y: pole 1 used (COM 3, NC 2, NO 4), pole 2 unused. Coils boot de-energized (gate pull-downs).

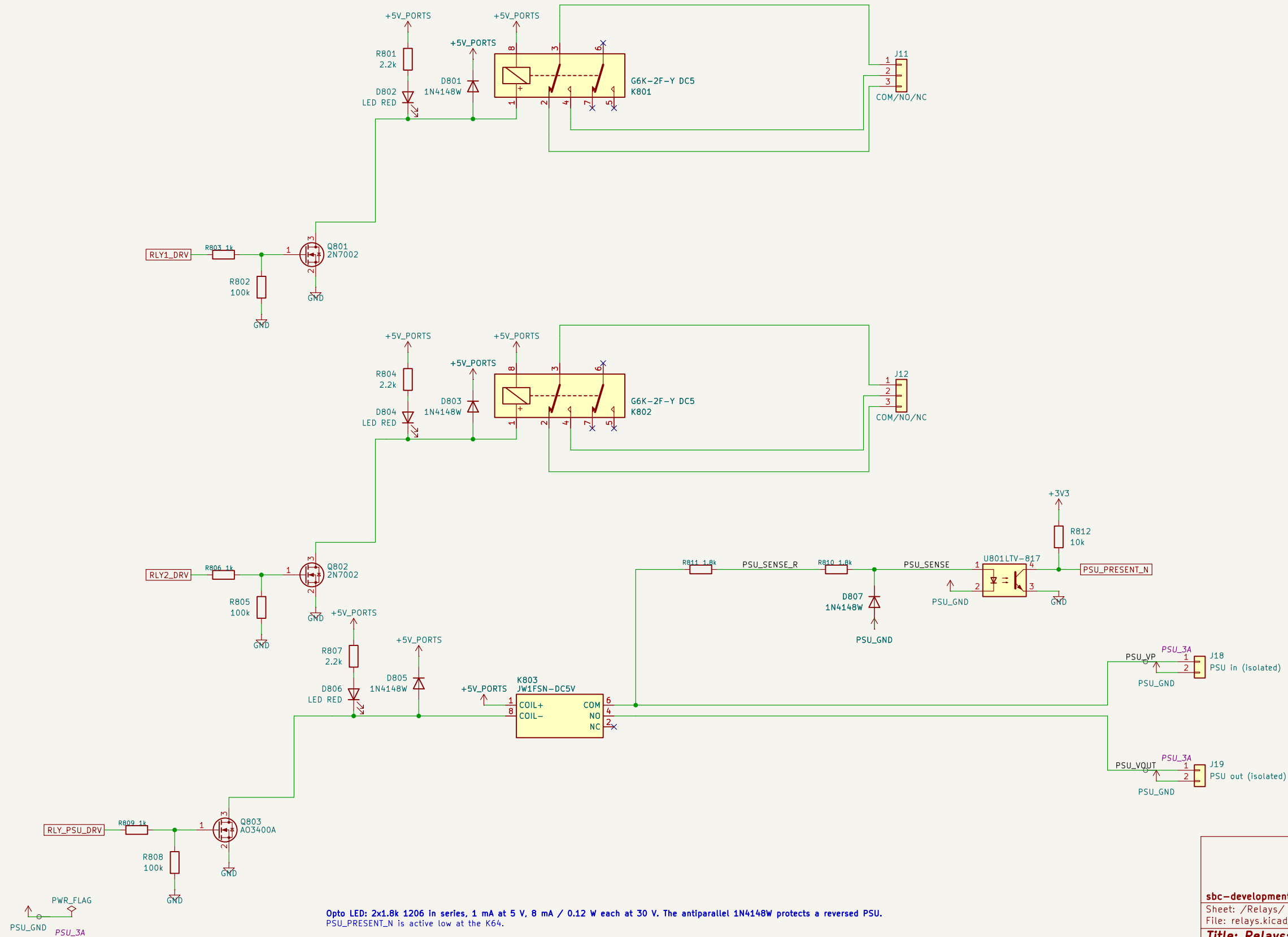
Red LEDs sit across each coil and show the energized state. Flyback 1N4148W on every coil.

Passthrough J18 -> K803 (NO contact) -> J19 on nets PSU_VP / PSU_VOUT / PSU_GND that touch no board net: PSU_GND is its own ground.

Only the optocoupler's emitter sits across the passthrough, sized for 5–30 V. No indicator LED there.

K803 JW1FSN pad map COM=6 NO=4 NC=2: VERIFY against the Panasonic terminal drawing before fab.

Current: the passthrough J18 -> K803 -> J19 (PSU_VP, PSU_VOUT, PSU_GND) is laid out for 3 A continuous: class PSU_3A, 2 mm traces or pours, two vias per layer change.



sbc-development-baseboard

Sheet: /Relays/
File: relays.kicad_sch

Title: Relays: two SPDT signal relays, isolated PSU passthrough

Size: A3	Date: 2026-10-03
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Rev: 0.1

KiCad E.D.A. 10.0.6

Id: 9/9